IMPLICATIONS OF EPISTEMIC MONOPOLY IN FRONTIER AI SYSTEMS: SYSTEMIC DEPENDECY, REGULATORY CAPTURE AND SINGLE POINTS OF FAILURE.

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Introduction

Epistemic monopoly in frontier AI models refers to institutional concentration of knowledge production, model development, infrastructure deployment and data availability by specific AI system providers resulting in systemic dependencies, regulatory capture and risks of exposure to single points of failure in critical systems.

Practical statistics and reports indicate monopoly at various AI ecosystems. Ranging from access and user prevalence, infrastructure concentration and economic dominance.

1. Access/User prevalence

As of 2026, AI user statistics report shows that ChatGPT has surpassed 800 million weekly active users as of early 2026, up from 400 million in February 2025. That doubling happened in under twelve months.[1] Google is reported to have made $1.2 billion from Gemini subscriptions in 2025. By early 2026, reports indicated 750 million monthly active users for the Gemini app, 2 billion monthly users for AI Overviews, and more than 8 million paid seats of Gemini Enterprise across 2,800 companies. Google reported Gemini had 750 million active users, mostly from people using AI overviews. Google Gemini has been downloaded over 430 million times since it launched in May 2024 Gemini had over 1 billion web visits in the fourth quarter of 2025.Google also reported **13 million developers** have built with its generative models.[2] Anthropic economic Index report Claude AI now has 18.9 million monthly active users on the web browser app, while the mobile app has 7.38 million monthly active users.

2. Economic dominance

OpenAI projected $29.4 billion in total revenue for 2026, up from an estimated $3.7 billion in 2024 and approximately $13 billion in 2025. That represents roughly 8x revenue growth over two

[1] Report by AI business weekly on Open AI Statistics 2026: Revenue, Users & Market Share accessed at :https://aibusinessweekly.net/p/openai-statistics#open-ai-user-statistics

[2] As Reported by business of Apps accessed at https://www.businessofapps.com/data/google-gemini-statistics and Panto at https://www.getpanto.ai/blog/google-gemini-statistics

years. Claude is currently valued at $380 billion and holds 4.5% of the AI chatbot market share.[3]Google is reported to have made $1.2 billion from Gemini subscriptions in 2025. Gemini had over 1 billion web visits in the fourth quarter of 2025.Google also reported **13 million developers** have built with its generative models.[4]

As of 2024, the top five frontier tech companies (Google, Amazon, Microsoft, Apple, Meta) were reported to command a combined $12 trillion in market capitalization and controlled a majority of the AI value chain

3. Infrastructure concentration

NVIDIA is reported to hold approximately 80-95% market share in AI chips, while three cloud providers (AWS, Azure, GCP) control 68-70% of the infrastructure required to train frontier models.[5]This would therefore mean that majority if not all AI companies rely on the same foundational infrastructure.

Translation to systemic Risk

Systemic dependency:

Systemic dependency refers to core reliance on prevailing AI systems by institutions, people and governments to solve or advance economic, social and institutional functions potentially resulting in increased widespread risks should such systems fail. Epistemic monopoly contributes to systemic dependency by eroding alternatives of model development, data access and infrastructure availability; limiting it only to the prevalent systems which in turn increase systemic risks in the event of failure.

Regulatory capture

Regulators rely on the dominant frontier system providers for technical information that informs development of policies, rules and laws intended to regulate the systems. Epistemic monopoly manifests when dominant frontier firms leverage their knowledge of technical expertise to shape regulations that apply to them, weakening regulatory autonomy. Market concentration by the prevalent frontiers also means monopolistic lobbying that could result in governance risk of industry self regulation as opposed to oversight.

[3] Report at https://www.demandsage.com/claude-ai-statistics/#:~

[4] As Reported by business of Apps accessed at https://www.businessofapps.com/data/google-gemini-statistics and Panto at https://www.getpanto.ai/blog/google-gemini-statistics

[5] Report accessed at: https://ea-crux-project.vercel.app/ai-transition-model/ai-control-concentration/

Single points of failure.

Single point of failure refers to the risk of dominant system breakdown, triggering cascading failures across dependent systems. Epistemic monopoly implies that dominant Frontier systems are primarily relied upon for core functionality by dependent systems-without much alternatives. In the event of failure in the primary infrastructure, the entire ecosystem is disrupted, which may result in widespread systemic loss.

Measuring dependency; The dependency index framework as a Governance tool.

As part of systemic dependency risk evaluation, this paper introduces a model dependency mapping tool to effectively identify and address the risks of frontier dependencies and development of a dependency index that would a step towards developing a tool that evaluates concentration in frontier systems. The Index provides a foundational step toward the development of a structured framework for monitoring and managing systemic dependency at large. This maybe be adopted by regulators as part of risk mapping and audit processes for oversight in deployment of frontier systems.

Dependency Index-Prototype V.01

Dimension	Implication	Indicator	Risk score/level	Resilience adjustment
Infrastructure dependency	Heavy reliance on concentrate compute and cloud systems.	Lack of alternatives, geographical location	Low, high, moderate	Availability of alternatives
Model dependency	Reliance on frontier AI models for core functions	Use of common AI models, lack of integration mechanisms,	Low, High, moderate	Interoperability mechanisms/ability to switch to models

Institutional dependency	Integration of AI into governance and societal systems Governance	Use in government, finance, health, justice Dependency of regulators on Frontier Ai for knowledge, technical expertise	Low, High, moderate Low, High, moderate	Operational continuity plan External advisory diversity. Independent technical expertise.

4. Proposed Interventions

Mitigating systemic dependency requires detailed identification of such dependencies as a first step; auditing efficiency of such models and mapping to what extent of dependency on particular frontier systems or models followed by decentralization of the same across systems to ensure the same does not concentrate in a single system provider.

To minimize regulatory capture and maintain autonomy, regulators should conduct independent oversight by implementing sovereign audits independent of dominant providers' standards. This maybe achieved through rigorous capacity building of independent experts and embedding specialized audit units within regulatory bodies tasked with conducting compliance checks, assessing system risks, authenticating safety measures and developing regulatory guidelines/standards separate of industry benchmarks.

Cases of single point failures can be avoided by implementing use of diverse foundational infrastructure to ensure that failure on one system does not cascade to the entire ecosystem. Developing continuity measures such as redundancy protocols and aiming towards diversification of infrastructure in the long term.